

Demonstration of Multi-Layered Inflatables: Asymmetric-Shaped Inflatable Structures by Stacked Planar Sheets

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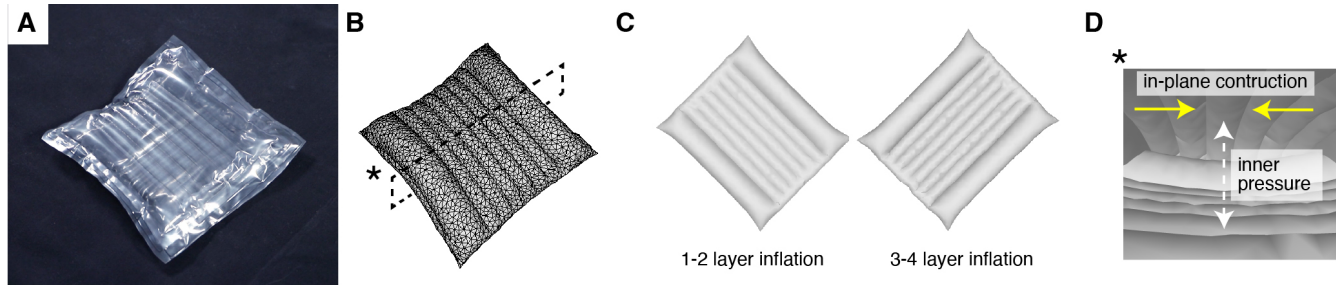


Figure 1: Overview of the Multi-Layered Inflatables. (A) Four planar sheets are stacked and bonded in pairs to form two external layers (sheets 1–2 and 3–4) and one internal layer (between sheets 2 and 3) and the layers employ (B) Three distinct air chambers. (C) The bonding patterns in each chamber are designed to induce different in-plane contraction behaviors during inflation, resulting in complex, asymmetric shapes. (D) The internal chamber, with its independent pressure control, enhances the stiffness of the external layers.

Abstract

We present “Multi-Layered Inflatables, a novel multi-layered inflatable structure that enables the design and fabrication of complex, asymmetric 3D forms from stacked planar sheets. Our approach employs some thin sheets that are assembled with distinct bonding patterns to form three separate air chambers. Two outer pairs form the external surfaces, while the intermediate chamber serves as an internal layer that reinforces the structure against external forces. By controlling the bonding patterns and independently tuning the pressure in each chamber using a low boiling point liquid (e.g., Novec), we achieve localized in-plane contraction and expansion. This allows us to generate forms such as dome-and-flat hybrids or other non-uniform surfaces—configurations that were previously challenging to obtain with single-layer or conventional methods.

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Additionally, the internal chamber increases the overall stiffness by counteracting the deformation of the external surfaces.

Keywords

Inflatable Structure, Digital Fabrication

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1 Introduction

Inflatable structures are attractive for applications in architecture [Ren et al. 2024], furniture [Sareen et al. 2017], and soft robotics [Niiyama et al. 2015] due to their lightweight and deployable nature. However, conventional methods typically yield simple, symmetric dome or shell shapes [Ren et al. 2024] and require complex joining of three-dimensional patches [Skouras et al. 2014], which poses challenges for rapid prototyping and mass production.

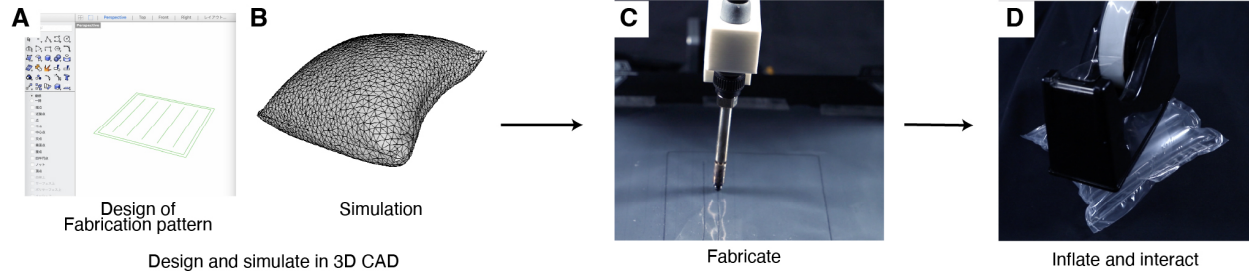


Figure 2: The hands-on scenarios. (A) First, attendees can design the fabrication patterns. (B) They quickly verify the shapes and modify the patterns using simulation. Then, they can (C) fabricate and (D) demonstrate their own-designed pattern.

In contrast, Multi-Layered Inflatables leverage stacked planar sheets with independently controlled bonding patterns to form distinct air chambers. The outer layers (formed by sheets 1–2 and 3–4 in Fig. 1) create the visible surfaces, while the inner layer (between sheets 2 and 3 in Fig. 1) provides reinforcement by mitigating deformation from external forces. By tailoring the bonding patterns, we can induce localized in-plane contraction and expansion upon inflation. This enables the creation of complex, non-uniform shapes (e.g., a dome-shaped upper surface combined with a flat base) that were previously difficult to achieve.

2 Methodology

Our structure is assembled from planar sheets arranged in the following manner; the outer layers (consisting of sheets 1–2 and sheets 3–4 in Fig. 1) form the upper and lower external surface, and the inner layers (consisting of sheets 2–3 in Fig. 1) reinforce the overall structure. Distinct air chambers are generated by the bonding patterns that leave unbonded regions between the sheets. Each chamber can be independently filled with a low-boiling point liquid (e.g., Novec) or air, allowing separate control of the pressure in each chamber. The bonding pattern in each chamber is designed to induce specific in-plane contraction behavior during inflation.

The advantages of this multi-layered design are twofold, complex shape formation and stiffness enhancement. By applying different bonding patterns in each layers localized in-plane contraction during inflation varies across the structure. For instance, one can design a structure where the upper external surface forms a dome while the lower external surface remains flat, or any other non-symmetric shape. This design freedom was not achievable using conventional single-layer or symmetric bonding methods. The internal chamber can be filled with a controlled amount of fluid, generating an independent pressure that helps resist deformations of the external layers from external loads. This internal pressure acts as a reinforcing cushion, improving the overall rigidity of the structure.

3 Design and Fabrication

We outline the following six-step process for designing and fabricating the multi-layered inflatable structure:

- (1) Using 3D CAD software, design the bonding and cutting patterns for each of the sheets.
- (2) Bond sheets according to the designed pattern to form the upper external layer.
- (3) Bond sheets similarly form the lower external layer.
- (4) Assemble the four sheets by aligning them so that the unbonded regions form the three air chambers.

- (5) Inject low-boiling point liquid into each chamber through dedicated ports to achieve the desired pressure profile. The pressure in the internal layer is tuned independently to reinforce the structure.
- (6) Inflate the assembled structure and observe the emergence of the complex, asymmetric shape along with enhanced stiffness due to the internal pressure.

For design, we use Rhinoceros and Grasshopper (Robert McNeel & Associates) as our 3D CAD software. For fabrication, we use a plastic film (Nylon Poly B type, FukuSuke Industries) for the sheets, a soldering-iron-modified (FX-888D, Hakko) Ender 3 (Creality) for bonding, and a CAMEO 5 (Silhouette Design) cutting plotter for precision cutting.

4 Demonstration Procedure

In the demonstration at Siggraph, participants will design and fabricate various examples of Multi-Layered Inflatables. At first, they will design the welding and cutting patterns using 3D CAD. By using the implemented Grasshopper plugin, they can easily generate the welding pattern (Fig. 2a). Through the numerical simulation (Fig. 2b), they quickly verify the inflated shapes of the designed models and modify the patterns. Following the procedure explained in Section 3, the attendees finally fabricate their models and inflate them using a hot plate or an air pump. Through our demonstration, they can observe the relationships between the designed patterns and the deformed shapes of the Multi-Layered Inflatables.

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